AI and ML Use in HireSense

Detailed Guide to Models, Training, Prompts, Inference, and Integration

What AI is used for

1. What AI and ML Mean

Artificial Intelligence (AI)

AI is software that can read, understand, and produce human-like text. In HireSense, AI helps analyse resumes, generate and evaluate interview questions, explain job matches, and coach English.

Machine Learning (ML)

ML is a branch of AI where software learns patterns from examples. A traditional ML model (small) can be trained on a dataset. A large language model is also learned, but we usually use one that someone else already trained.

Our simple rule

Ask: did we change the model's weights using our own dataset? If yes, that is training. If no, we are only using a pretrained model or ordinary code.

AI vs ML vs rules

- AI feature: uses a language model to read/write text (resume analysis, question generation).

- ML

2. What AI and ML Components Exist

Component	Type	Where used
Ollama qwen3:4b	Pretrained local LLM	Resume analysis, interview Q/A/evaluation, English coach, match explanation
Sentence Transformers all-MiniLM-L6-v2	Pretrained local embedding model	Job matching, duplicate question detection
TF-IDF + logistic regression classifier	Our own trained ML model	Question-type cross-check

Kokoro-82M TTS	Pretrained local speech model	Live voice interview speech
Whisper base.en STT	Pretrained local speech model	Live voice transcription
MediaPipe Face Landmarker	Pretrained browser vision model	Camera monitoring

3. Detailed AI and ML Model-by-Model Explanation

A. Ollama local LLM: qwen3:4b

- Model name/family: qwen3:4b. A 4-billion parameter general-purpose language model.

- Pro

B. Sentence Transformers: all-MiniLM-L6-v2

- Model: sentence-transformers/all-MiniLM-L6-v2.

- Ty

C. Our own ML model: question-type classifier

- Task: predict question type (technical, hr, behavioral, communication, project).

- D
admin

D. Kokoro-82M and Whisper base.en (speech)

- TTS: Kokoro-82M local model for Aria's voice (~330 MB), uses espeak-ng phonemes (GPL-3.0).

- ST

E. MediaPipe Face Landmarker (vision)

- Model: MediaPipe Face Landmarker running in the browser.

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4. Does Our Project Use a Trained Model?

Short answer: We trained one small traditional ML model. We did NOT train or fine-tune the large AI models. Those are pretrained models that we download and run locally.

Question	Answer
Did we train a language model from scratch?	No; qwen3:4b is downloaded and run.
Did we fine-tune qwen3:4b?	No evidence; none found.
Did we train the embedding model?	No; all-MiniLM-L6-v2 is pretrained.
Did we train the STT/TTS/vision models?	No.
Did we train any ML model?	Yes; the question-type classifier (scikit-learn).
Can training evidence be shown?	Yes via backend/ml/train_question_classifier.py and README.

5. How We Use AI in Simple Steps

A. Resume analysis workflow

```
Student uploads PDF/DOCX
```

-> f

B. Interview question generation workflow

```
Start interview request
```

-> i

C. Answer evaluation workflow

```
Student submits answer
```

```
-> i
```

D. Job matching workflow

```
Student applies for job
-> m
```

E. English practice workflow

```
Student sends message
-> e
```

6. Prompt Engineering

We do not train the model for each feature. Instead we send carefully designed instructions called prompts. A prompt includes context (resume facts, previous answers, role) and asks the model to return structured JSON. The backend validates that output. Prompt engineering means designing and testing these instructions.

- Resume text is wrapped in <resume> tags and the model is told to ignore instructions inside it (prompt-injection protection).

- Ea

7. Why We Chose Local AI

- Privacy: resume text stays on the student's computer.

- No

8. Limitations of AI and ML in HireSense

- The 4B LLM can be generous, can repeat questions, and can ignore some instructions.

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9. Evidence from Code

- backend/app/services/ai.py - prompts, schemas, validation, limits, logging.

- ba

10. Simple Explanation for Viva

For most features, we are using pretrained AI through prompts and validation rules. For one feature, we trained a small scikit-learn model on a hand-written dataset. This is honest engineering: AI helps, but rules and checks keep results reliable.